

Research Proposal

Lin Weiye
2026/05/18

Research Theme

State-Locked Decoupling of Photoinduced Halide Phase Segregation in Wide-Bandgap Mixed-Halide Perovskites

Abstract

Wide-bandgap (1.65–1.80 eV) mixed-halide perovskites are key top absorbers for perovskite/silicon tandem solar cells, but their voltage and spectral stability are limited by photoinduced I/Br phase segregation, which causes quasi-Fermi-level-splitting reduction and open-circuit voltage loss. This project proposes a **state-locked decoupling framework** that experimentally and theoretically separates **reversible optical readout** (PL, KPFM) from **irreversible chemical / structural memory** (an auxiliary field q): drive the material into reproducible trajectory checkpoints S_0 – S_5 , suppress further evolution by low-temperature locking, and read each locked state using optical, chemical, structural, electrostatic, and defect-sensitive probes. The locked-state fingerprints will constrain a **driven** Cahn-Hilliard composition-evolution model (with non-equilibrium carrier coupling and an irreversibility channel) and identify which physical term dominates onset, growth, saturation, and recovery. The expected result is a falsifiable mechanism map, with a hard pre-registered failure condition, rather than a PL-only description of phase segregation.

1. Background and Research Question

Wide-bandgap mixed-halide perovskites with bandgaps of about 1.65–1.80 eV are essential for high-efficiency tandem photovoltaics⁸. Their adjustable I/Br ratio enables bandgap control, but the same compositional flexibility also creates a serious instability: under illumination, initially mixed halides can segregate into iodide-rich and bromide-rich regions¹. The iodide-rich regions have lower bandgap, collect carriers by funneling, red-shift photoluminescence (PL), and reduce quasi-Fermi-level splitting and open-circuit voltage^{1,5}.

Previous studies have clarified important parts of this process. Hoke et al. established reversible red-shifted emission in mixed-halide perovskites¹; Brivio et al. described the thermodynamic miscibility gap²; Tang³, Zhao⁴, Peng⁶, and Fan⁷ showed that local microstructure, strain, and cryogenic imaging reveal spatially heterogeneous segregation pathways. A unified theoretical account that combines thermodynamics, electrostatics, and carrier funneling has also been proposed⁵. However, the field still lacks an operational method to decide whether different measurements are observing the same material state. A PL time point is not enough, because PL can be dominated by a small low-bandgap domain and may not represent average composition or complete recovery.

The central research question is:

Can photoinduced halide phase segregation be divided into reproducible operational states, and can locked-state readout identify which physical term controls each transition?

This question is motivated by four unresolved contradictions in the literature: (C1) **where** segregation starts (grain boundary, grain interior, or intragrain clusters); (C2) **when** it truly begins (threshold crossing, classical nucleation, or spinodal amplification, with possible non-optical precursors preceding PL onset); (C3) **what** PL actually reports (chemical composition or funneling-weighted emission from a small low-bandgap domain); and (C4) **whether** optical recovery implies material recovery. The proposed project therefore shifts the unit of analysis from a single PL trace to a **locked material state**, so that each contradiction is tested against state-resolved, multi-modal endpoints rather than a PL time point.

2. Objective, Hypothesis, and State Definition

The objective is to establish a state-resolved experimental and physical framework for photoinduced phase segregation in wide-bandgap mixed-halide perovskites. The framework must be useful even if the initial state boundaries are revised by data.

The central hypothesis is:

Photoinduced halide phase segregation can be segmented into reproducible operational states S_0 to S_5 . The conserved halide composition field $c(\mathbf{r}, t)$ evolves according to a Cahn-Hilliard-type framework, while mobility, electrostatic, elastic, interface, and memory channels determine which term dominates each state transition.

The proposed operational states are:

Each state is defined by a **primary** trigger (a numerical threshold on a single anchor observable) and **secondary** corroborating channels. When a secondary channel disagrees with the primary trigger, the disagreement itself is recorded as a state-resolved diagnostic rather than being averaged away. Numerical thresholds are to be locked at the end of Year 1 from baseline statistics.

State	Operational meaning	Primary trigger (anchor)	Secondary / corroborating channels
S_0	dark mixed baseline	all channels within $\pm 2\sigma$ of dark baseline	—
S_1	photoactivated precursor	KPFM Δ CPD > threshold <i>before</i> PL red-fraction exceeds noise floor	SPV, weak operando XRD shift
S_2	segregation onset	PL red-fraction crosses threshold (numerical, to be set in Year 1)	GIWAXS/XRD peak splitting, SIMS/EELS contrast
S_3	growth	PL red-fraction increases monotonically with dose	spatial domain count, GIWAXS peak area
S_4	saturated segregated state	PL red-fraction plateau ($d/dt < \text{threshold}$)	at least one non-PL channel also plateaus
S_5	recovery / memory	PL recovers to $\leq 110\%$ of S_0 baseline after dark relaxation	residual $q(\mathbf{r})$ in XPS / SIMS / GIWAXS PbI_2 tail

The measured fingerprint of state S_k is:

$$Z(S_k) = [c_k(\mathbf{r}), \bar{v}_k, \phi_k(\mathbf{r}), \varepsilon_k(\mathbf{r}), q_k(\mathbf{r})] \quad (1)$$

Here $c(\mathbf{r})$ is the local iodide fraction, $\bar{v}$ is a **sample-averaged scalar** ionic-conductivity parameter (admittance/impedance derived; the only non-field entry in Z), $\phi(\mathbf{r})$ is a surface/local potential proxy, $\varepsilon(\mathbf{r})$ is strain or microstrain, and $q(\mathbf{r})$ is the auxiliary field for irreversible chemistry / trapped-charge memory. S_k are therefore **reproducible trajectory checkpoints along a controlled protocol**, not thermodynamic states; the framework admits hysteresis and non-Markovian channels through q . The fingerprint is not claimed to be a complete microscopic state — it is an experimentally accessible constraint on the model.

3. Physical Framework

The primary order parameter is the local iodide fraction. Within the segregation regime (no halide loss to side phases), it is conserved:

$$c(\mathbf{r}, t) = x_1(\mathbf{r}, t), \quad d/dt \int_{\Omega} c(\mathbf{r}, t) dV = 0 \quad (\text{when } R_q = 0) \quad (2)$$

For a mixed I/Br lattice, a regular-solution-type free energy gives the basic demixing tendency:

$$f_{\text{mix}}(c) = (k_B T / \Omega_0) [c \ln c + (1 - c) \ln(1 - c) + \chi c(1 - c)] \quad (3)$$

The working effective free-energy functional is:

$$\mathcal{G}[c] = \int_{\Omega} [f_{\text{mix}}(c) + (\kappa/2)|\nabla c|^2 + f_{\text{carrier}}(c, n, p) + f_{\text{elastic}} + f_{\text{elec}}] dV + G_{\text{interface}} \quad (4)$$

Here f_{carrier} is **not** a thermodynamic free energy: it is the effective potential obtained after adiabatic elimination of fast photo-excited carriers, expressed through their quasi-Fermi levels⁵; illumination enters as a non-equilibrium driver rather than a state function. The corresponding driven Cahn-Hilliard-type evolution equation⁹ is:

$$\partial c / \partial t = \nabla \cdot [M(c, \bar{v}, T) \nabla \mu_c] - k(c, T) c + k_{\text{rev}} q, \quad \partial q / \partial t = k(c, T) c - k_{\text{rev}} q, \quad \mu_c = \delta G / \delta c \quad (5)$$

Eq. (5) is closed by no-flux boundary conditions at the film/substrate and air interfaces; $G_{\text{interface}}$ supplies the grain-boundary and electrode surface-exchange terms. Variables: $\bar{v}$ is a **sample-averaged** ionic-conductivity parameter calibrated from impedance/admittance (not a spatial field, in contrast to $c(\mathbf{r})$ and $q(\mathbf{r})$); $q(\mathbf{r}, t)$ is an auxiliary field tracking irreversible chemistry / trapped-charge memory (e.g., halide bound to a secondary $\text{PbI}_2/\text{PbBr}_2$ phase, deep-trap occupation); k and k_{rev} are forward/reverse rate coefficients. The pair (Eq. 5a, Eq. 5b) conserves the total halide budget $c + q$; the conservation in Eq. (2) holds whenever k can be set to zero by experimental design (e.g., inert atmosphere, sub-degradation light dose).

In this formulation, f_{mix} and the gradient term describe thermodynamic demixing and domain formation; $M(c, \bar{v}, T)$ describes vacancy-assisted ionic mobility; f_{elec} and f_{elastic} capture potential and strain contributions; f_{carrier} is the non-equilibrium driver; k selects the irreversible-chemistry channel for state S_5 memory. The model is used diagnostically: for each $S_k \rightarrow S_{k+1}$ transition, the dominant term is the one that best explains onset position, growth rate, saturation, or residual memory with stated uncertainty.

4. Methodology

The experimental strategy has four steps. First, samples will be driven into target states by controlled light dose, wavelength, intensity, temperature, and atmosphere. Operando PL hyperspectral imaging will track peak position, red

fraction, linewidth, and spatial contrast. KPFM or SPV will be used where possible to detect potential-related precursors.

Second, once a target S_k is reached, further evolution will be suppressed by low-temperature state preservation. Practical routes include cryostage quenching to 170-100 K, liquid-nitrogen cold-stage transfer near 77 K, and plunge-freeze cryo preparation for TEM-compatible samples when collaboration is available. Locking will be validated by delay series, dark-transfer controls, low-temperature PL, beam-dose tests, and sister-sample replication. A successful lock means that drift during transfer/readout is smaller than state-to-state contrast or is explicitly included as uncertainty.

Third, the locked state will be read by scale-tagged endpoint characterization:

Target	Main readout	Model role
c	EELS, TOF-SIMS, XPS, GIWAXS/XRD	composition, mixing, gradient terms
$\bar{v}$ (scalar)	admittance/impedance (primary), DLTS / TRPL / hysteresis (secondary, calibration only)	mobility modifier $M(c, \bar{v}, T)$
ϕ	KPFM, SPV	electrostatic or surface-potential contribution
ϵ	GIWAXS, XRD, Raman, 4D-STEM	elastic contribution
q	XPS, SIMS, PbI ₂ -related XRD, recovery-delay tests	irreversible chemistry or memory

Fourth, the measured fields will be mapped to model terms. The core publishable path is PL/KPFM or SPV plus XPS/SIMS and GIWAXS/Raman. Cryo-CL, cryo-EELS, 4D-STEM, and cryo-STEM-EDS are validation paths, not requirements for project success.

5. Three-Year Research Plan

Period	Main work	Acceptance gate
Year 1	4-point composition pre-screen ($\text{Cs}_x\text{FA}_{1-x}\text{Pb}(\text{I}_{1-y}\text{Br}_y)_3$, $y \in \{0.20, 0.27, 0.33, 0.40\}$) by operando PL to select one baseline composition; build fiducial-marked thin films on glass; establish operando PL hyperspectral protocol + one non-optical channel (KPFM or impedance for $\bar{v}$); develop single cryogenic locking protocol (170 K cryostage) with a quench-time vs ionic-diffusion-time (Damköhler) validation; lock numerical thresholds for S_0, S_2, S_4 from baseline statistics. (77 K plunge-freeze and device-like stacks deferred to Year 2.)	Locking fidelity is quantified by a Damköhler-number argument (cooling time $\ll$ fastest mode); transfer/readout drift is smaller than state contrast or reported as uncertainty; quantitative S_k trigger table published.
Year 2	Extend trigger table to S_1, S_3, S_5 ; add scale-tagged endpoint channels (XPS/SIMS chemical; GIWAXS/Raman structural); correlate fiducial-matched ROI-statistics across optical, chemical, and structural channels; introduce 77 K plunge-freeze for selected states with collaborators if available; build mini-devices on the baseline composition to track quasi-Fermi-level splitting (QFLS) / V_{oc} retention as a PV-relevant bridge metric.	At least two independent endpoint channels correlate with state labels at $> 2\sigma$; QFLS retention is reported per state; Paper 1 (state-locked decoupling) submitted.
Year 3	Analyze $S_4 \rightarrow S_5$ recovery and memory; fit measured $Z(S_k)$ changes against Eq. (5) terms; construct a transition-term map for onset / growth / saturation / recovery on the baseline composition (regime-diagram-style extension across y only if Year 2 yields tight error bars).	The model improves on a PL-only kinetic description against a pre-registered metric, or the state boundaries are revised; Paper 2 and thesis chapters completed.

6. Expected Outcomes, Originality, and Risk Management

The expected outcomes are: (1) reproducible operational definitions for S_0 - S_5 ; (2) a transferable write-lock-read-map protocol; (3) a state-resolved mechanism map assigning dominant physical terms to transitions; and (4) two publication-ready datasets, one focused on state-locked decoupling and one on the mechanism regime diagram.

The originality lies in three aspects. First, the project treats phase segregation as a state-resolved physical process rather than a PL red-shift alone. Second, locking is triggered by physical state criteria rather than arbitrary illumination time. Third, the model explicitly separates conserved composition evolution from sample-averaged ionic-conductivity, proxy variables, and irreversible memory, avoiding overclaiming complete microscopic inversion. The term **"decoupling"** in this proposal denotes the experimental and modelling separation of (i) **reversible optical readout (PL, KPFM)** from (ii) **irreversible chemical / structural memory (the auxiliary field $q(\mathbf{r})$ tracked by XPS, SIMS, PbI₂-XRD)**, anchored at each locked state.

The main risks and fallback strategies are:

Risk	Fallback
State locking is imperfect	quantify drift via Damköhler-number test and include residual as uncertainty
Cooling-induced artifacts (thermal contraction, low-T PL line-shape, exciton-polaron renormalisation)	report low-T endpoints with a paired unilluminated cooled-only control sample to subtract baseline shifts
Same-statistics ROI registration fails	fall back to fiducial-matched statistical regions (not pixel-level same-ROI) and ROI-level distributions
KPFM/SPV is surface-sensitive	report ϕ only as a surface/local potential proxy
SIMS/EELS damages the sample	use dose series and cross-check with XPS/GIWAXS
Cryogenic microscopy collaboration is unavailable	retain PL/KPFM(SPV)/XPS/SIMS/GIWAXS/Raman as the core path
S_0 - S_5 is incomplete	revise state boundaries based on data rather than forcing the initial model

Overall, the project is designed to remain falsifiable. A **hard failure condition** is committed in advance: if, by the end of Year 2, no non-PL endpoint channel correlates with the PL-defined state labels at $> 2\sigma$ across replicates, the state-locked premise is abandoned and the work reverts to a kinetic-PL-only framework. Beyond this, soft failures are absorbed by uncertainty: if a channel fails, the framework is reduced to validated variables; if the six-state model is wrong, the state definition is revised; if locking is imperfect, the uncertainty becomes part of the result.

References

1. Hoke, E. T.; Slotcavage, D. J.; Dohner, E. R.; Bowring, A. R.; Karunadasa, H. I.; McGehee, M. D. Reversible Photo-Induced Trap Formation in Mixed-Halide Hybrid Perovskites for Photovoltaics. *Chemical Science* **2015**, 6, 613-617.
2. Brivio, F.; Caetano, C.; Walsh, A. Thermodynamic Origin of Photoinstability in the $\text{CH}_3\text{NH}_3\text{Pb}(\text{I}_{1-x}\text{Br}_x)_3$ Hybrid Halide Perovskite Alloy. *Journal of Physical Chemistry Letters* **2016**.
3. Tang, X.; van den Berg, M. F. A.; Gu, E.; Horneber, A.; Matt, G. J.; Osvet, A.; Meixner, A. J.; Zhang, D.; Brabec, C. J. Local Observation of Phase Segregation in Mixed-Halide Perovskite. *Nano Letters* **2018**.
4. Zhao, Y.; Miao, P.; Elia, J.; Hu, H.; Wang, X.; Heumueller, T.; et al. Strain-Activated Light-Induced Halide Segregation in Mixed-Halide Perovskite Solids. *Nature Communications* **2020**, 11, 6328.
5. Chen, Z.; Brocks, G.; Tao, S.; Bobbert, P. A. Unified Theory for Light-Induced Halide Segregation in Mixed Halide Perovskites. *Nature Communications* **2021**, 12, 2687.
6. Peng, S.; Wang, Y.; Braun, M.; Dionne, J.; Cai, W.; McIntyre, P. C. Kinetics and Mechanism of Light-Induced Phase Separation in a Mixed-Halide Perovskite. *Matter* **2023**, 6, 2052-2065.
7. Fan, Q.; Cui, Y.; Li, Y.; Karunadasa, H. I.; Lindenberg, A. M.; et al. Phase Segregation Dynamics in Mixed-Halide Perovskites Revealed by Plunge-Freezing Cryo-Electron Microscopy. *Cell Reports Physical Science* **2025**, 6, 102653.
8. Nie, T.; Fang, Z.; Ren, X.; Duan, Y.; Liu, S. Recent Advances in Wide-Bandgap Organic-Inorganic Halide Perovskite Solar Cells and Tandem Application. *Nano-Micro Letters* **2023**, 15, 70.
9. Cahn, J. W.; Hilliard, J. E. Free Energy of a Nonuniform System. I. Interfacial Free Energy. *Journal of Chemical Physics* **1958**, 28, 258-267.